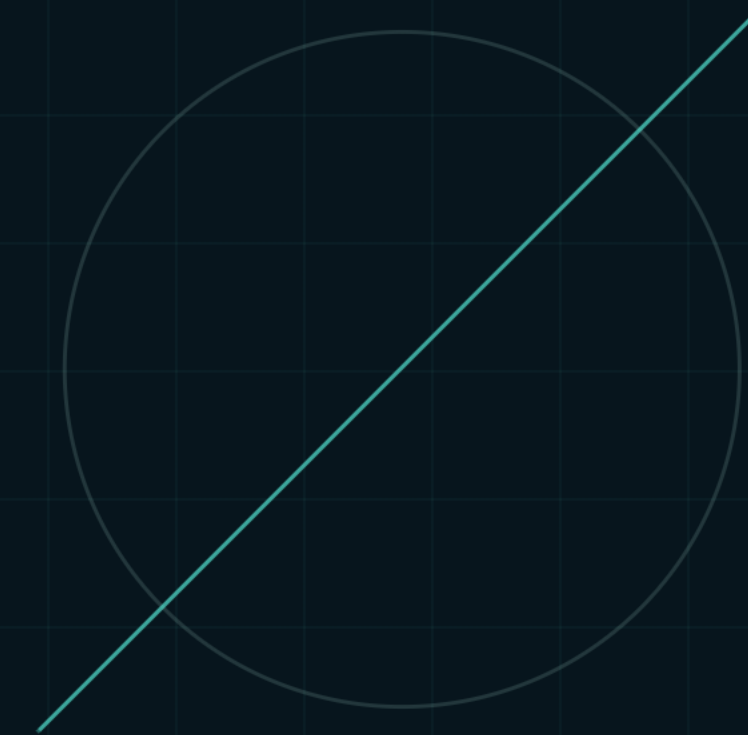


01 / EMBEDDED SYSTEMS

# ESP32 Hardware



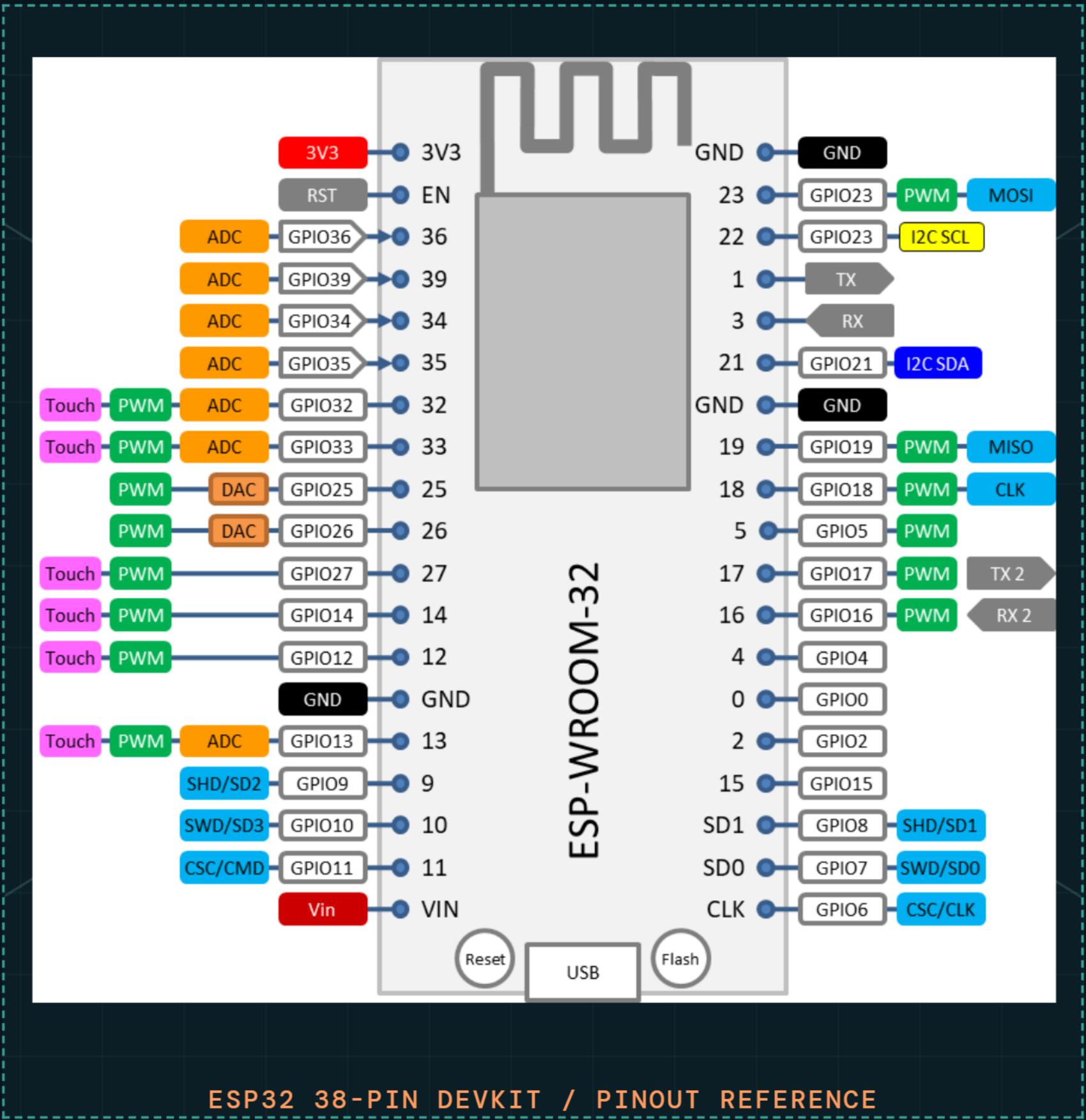
| A tiny board with the reach of a whole lab.

02 / IDENTIFY THE BOARD

# ESP32 38-pin breakout.

A development board built around the ESP32-WROOM-32 module, exposing 19 header positions on each side.

|        |            |          |
|--------|------------|----------|
| 38     | 3.3V       | 240      |
| HEADER | GPIO LOGIC | MHZ DUAL |
| PINS   |            | CORE     |



03 / THE 38-PIN LAYOUT

# Read the headers from the outside in.

Pin names vary slightly between manufacturers. Always compare the silkscreen on your board with its datasheet before wiring.



⚠️ Style reference. Header order and labels can vary by manufacturer; verify the silkscreen before wiring.

04 / COMPLETE SIGNAL REFERENCE

# What each group is good for.

POWER

VIN · 3V3 · GND

VIN accepts regulated 5 V through the board regulator. 3V3 is the regulated output. Connect grounds together before testing.

DIGITAL I/O

GPIO 0–33

General-purpose digital pins. Most support input, output, interrupts, and PWM. GPIO 34–39 are input-only.

ANALOG

ADC1: 32–39

Use ADC1 for analog sensors when Wi-Fi is active. ADC2 conflicts with Wi-Fi and is less dependable for that use.

SPECIAL

DAC · TOUCH · RTC

DAC output: GPIO25/26. Capacitive touch: GPIO0, 2, 4, 12–15, 27, 32, 33. RTC GPIOs support low-power wake.

05 / COMMUNICATION DEFAULTS

# One board, many buses.

| BUS              | DEFAULT PINS                             | TYPICAL HARDWARE       | NOTES                                          |
|------------------|------------------------------------------|------------------------|------------------------------------------------|
| I <sup>2</sup> C | GPIO 21 SDA<br>GPIO 22 SCL               | OLED, IMU, sensors     | Pull-up resistors are normally required.       |
| VSPI             | GPIO 23 MOSI<br>19 MISO · 18 SCLK · 5 CS | SD card, displays      | Chip select is software-controlled.            |
| UART2            | GPIO 17 TX2<br>GPIO 16 RX2               | GPS, motor controllers | Use a common ground and matching logic levels. |
| UART0            | GPIO 1 TX0<br>GPIO 3 RX0                 | USB serial monitor     | Shared with programming and debug.             |

06 / PIN MAP AT A GLANCE

# A quick-reference cheat sheet.

SAFE STARTERS

**GPIO 16, 17, 18, 19, 21, 22, 23, 25, 26, 27, 32, 33**  
Good general choices after checking your peripheral requirements.

FLASH RESERVED

**GPIO 6–11**  
Connected to the module’s SPI flash. Do not use them as breakout pins.

INPUT ONLY

**GPIO 34, 35, 36 / VP, 39 / VN**  
Read sensors here. They cannot drive an LED, relay, or motor.

BOOT SENSITIVE

**GPIO 0, 2, 5, 12, 15**  
External pull-ups or pull-downs can stop the board from booting.

## 07 / POWER AND PROTECTION

# Three rules before power-up.

-  Use 3.3 V logic  
ESP32 GPIO is not 5 V tolerant. Level-shift 5 V sensor signals.
-  Share the ground  
Every external module needs a common GND reference with the ESP32.
-  Feed motors separately  
Motor noise and current spikes belong on their own supply, with a driver between.

## CHECK THIS FIRST

**Never connect 5 V directly to a GPIO.**

Power the board through USB, VIN, or a clean regulated 5 V source. Confirm polarity and current capacity before attaching a battery.

5 V  3.3 V  GPIO

## 08 / FROM CODE TO MOTION

# Sense. Decide. Act.

A reliable embedded project keeps the loop simple: collect a signal, make a decision, produce a visible result.

GPIO → peripheral → logic → output

```
// read, decide, respond
const distance = readSensor();
if (distance < SAFE_LIMIT) {
  stopMotors();
  signalWarning();
} else {
  driveForward();
}
```



09 / THANK YOU

# Thank You.

Let's learn how to program and turn ideas into real hardware.

ESP32 · CODE · CREATE

LET'S  
PROGRAM

